Exoplanet Habitability Prediction Using Machine Learning

1. Introduction

The search for habitable exoplanets is one of the most important areas of modern astronomical research. With the increasing availability of planetary and stellar data, traditional analysis methods become inefficient. Machine Learning (ML) provides powerful techniques to analyze complex patterns in large datasets and predict the habitability of exoplanets effectively.

This project presents a complete end-to-end system for predicting exoplanet habitability using machine learning models, supported by data preprocessing, feature engineering, model evaluation, backend APIs, frontend visualization, and cloud deployment.

2. Objectives of the Project

The key objectives of this project are:

- To collect and analyze exoplanet datasets from open-source platforms
- To clean, preprocess, and engineer features relevant to habitability
- To handle missing data and class imbalance effectively
- To build and compare multiple machine learning models
- To predict and rank exoplanets based on habitability probability
- To develop a full-stack application for real-time prediction
- To deploy a production-ready system with visualization support

3. Dataset Collection and Understanding

Multiple datasets related to exoplanets and host stars were explored. These datasets contained planetary parameters such as radius, mass, temperature, orbital period, and stellar characteristics.

Key observations during dataset collection include:

- Presence of missing and incomplete values
- Lack of predefined habitability class labels in some datasets
- High imbalance between habitable and non-habitable samples
- Variation in data quality across different sources

To address the absence of labels, scientific threshold rules and domain knowledge were applied to generate habitability classes.

4. Data Cleaning and Feature Engineering

A comprehensive preprocessing pipeline was implemented to improve data quality and model performance.

Data Cleaning Steps

- Detection and imputation of missing values
- Removal of irrelevant and redundant features
- Correction of inconsistent data types
- Identification and treatment of outliers using boxplots and statistical measures

Feature Engineering Techniques

- Feature scaling using standardization
- Encoding categorical features using label encoding and one-hot encoding
- Correlation analysis for feature selection
- Handling class imbalance using SMOTE
- Creation of scientific indices such as:
    - Habitability Score Index (HSI)
    - Stellar Compatibility Index (SCI)

These steps ensured the dataset was clean, balanced, and suitable for modeling.

5. Dataset Preparation for Machine Learning

After preprocessing, the dataset was prepared for model training using standard ML practices.

Key preparation steps include:

- Defining the target variable as:
    - Binary habitability class (Habitable / Non-Habitable)
    - Multi-class habitability levels (Low, Medium, High)
- Selecting features based on correlation with habitability indicators
- Splitting data into training and testing sets using an 80:20 ratio
- Building preprocessing pipelines for consistent scaling and encoding

This approach ensured reliable evaluation and avoided data leakage.

6. Machine Learning Models and Evaluation

Several machine learning models were trained and evaluated to compare performance.

Models Implemented

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)
- Decision Tree
- Random Forest
- XGBoost

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- Confusion Matrix analysis

Key Findings

- Random Forest and XGBoost showed strong performance with high accuracy
- XGBoost handled class imbalance and complex feature interactions effectively
- Linear models performed reasonably but struggled with minority classes
- Ensemble models provided the most reliable predictions

XGBoost was selected as the final model for habitability ranking.

7. Exoplanet Ranking and Analysis

Using probability outputs from the trained XGBoost model:

- Exoplanets were ranked based on predicted habitability likelihood
- Higher-ranked planets indicate stronger potential for supporting life
- Results were validated to ensure consistency with astrophysical reasoning

This ranking system supports prioritization of exoplanets for further study.

8. Backend API Development

A Flask-based backend was developed to deploy the trained models and enable real-time predictions.

Key backend features include:

- REST API endpoints for accepting planetary and stellar inputs
- JSON-based request and response structure
- Integration of trained ML models for prediction
- Secure endpoints with authentication mechanisms
- Database connectivity for storing inputs and prediction results

The backend ensures smooth communication between user input and model output.

9. Frontend User Interface

A clean and responsive frontend was built using HTML, CSS, and Bootstrap.

Frontend capabilities include:

- Structured input forms for planetary and stellar parameters
- Dynamic display of habitability predictions and scores
- Ranking tables for comparing multiple exoplanets
- Clear visualization of important features influencing habitability

The interface was designed to be intuitive and user-friendly.

10. Visualization and Dashboard

An interactive dashboard was implemented to improve interpretability and analysis.

Dashboard components include:

- Feature importance visualizations
- Habitability score distribution plots
- Correlation analysis between star and planet parameters
- Export of analytical insights in PDF and Excel formats

These visual tools enhance data-driven decision-making and model transparency.

11. Deployment and Documentation

The complete system was deployed on a cloud platform with database integration.

Final deliverables include:

- Live deployed application
- Technical documentation and README
- Final project report and demo materials

The system is fully functional and ready for real-world usage.

12. Conclusion

This project successfully demonstrates the application of machine learning and full-stack development to solve a real-world scientific problem. By integrating data preprocessing, advanced ML models, REST APIs, interactive UI, visualization dashboards, and cloud deployment, the system provides a scalable and efficient solution for exoplanet habitability prediction.
The project reflects strong practical skills in data science, machine learning, backend and frontend development, and deployment, making it a comprehensive and impactful AI-based solution.